

“Cloud Computing with AWS – Week 3 Assignment”

INTRODUCTION

Cloud Computing = Cloud computing is a technology that allows users to access computing resources such as servers, storage, and databases over the internet instead of using local systems. It provides flexibility, scalability, and cost efficiency.

Amazon Web Services (AWS) = Amazon Web Services (AWS) is a leading cloud platform that offers a wide range of services including computing, networking, databases, and serverless computing. AWS helps organizations build and deploy applications easily without managing physical infrastructure.

ASSIGNMENT 1

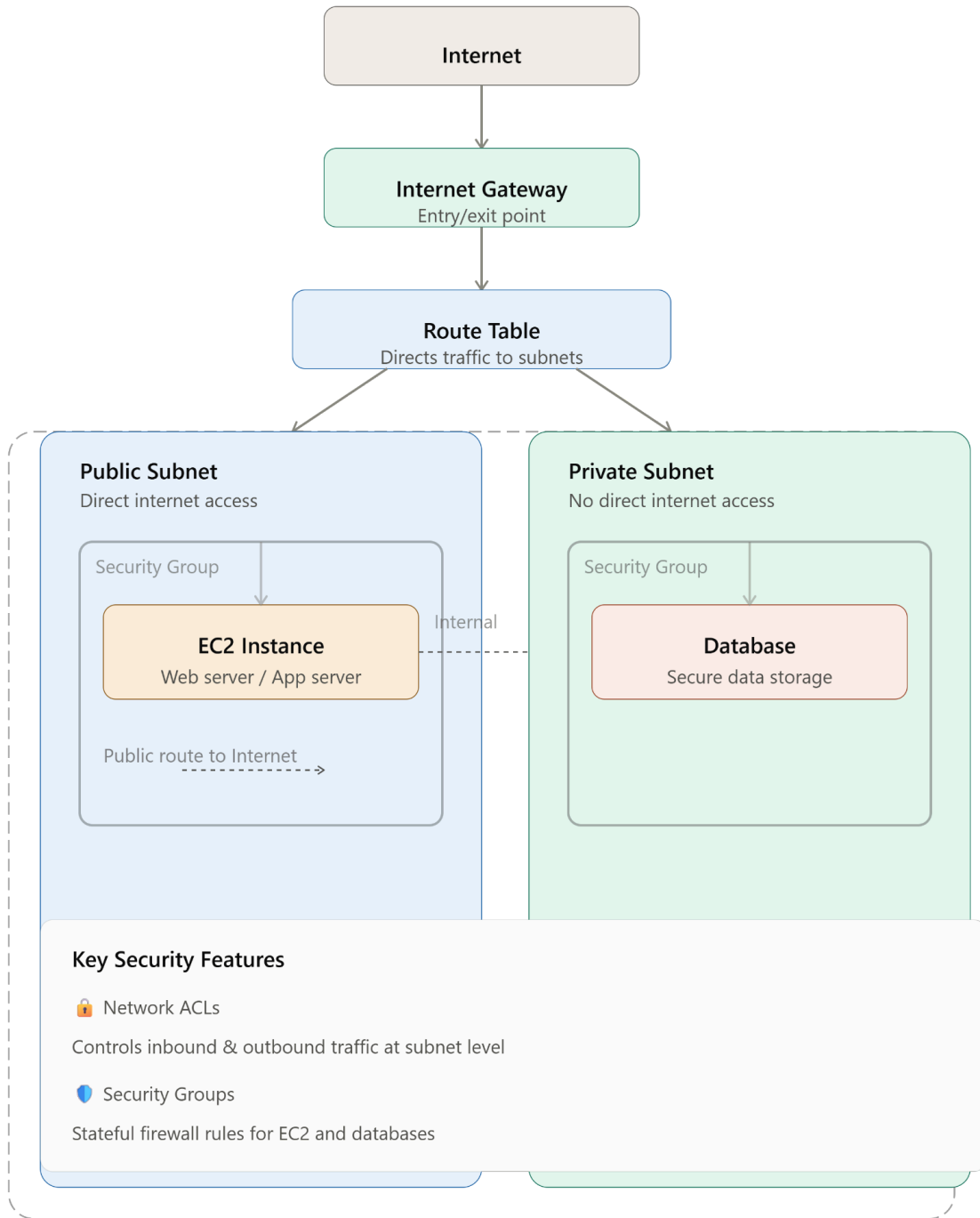
Setup a VPC with Public & Private Subnets

OBJECTIVE

The objective of this assignment is to understand how to create a Virtual Private Cloud (VPC) in AWS. A VPC allows users to create a secure and isolated network where resources like EC2 instances can be launched and managed safely.

EXPLANATION

A Virtual Private Cloud (VPC) is a logically isolated network in AWS. It enables users to define IP address ranges, create subnets, and control traffic using route tables and security groups. Public subnets allow internet access, while private subnets are used for secure internal resources like databases.



STEPS

- 1) Create a VPC with CIDR block 10.0.0.0/16
- 2) Create a public subnet and a private subnet
- 3) Attach an Internet Gateway to the VPC
- 4) Configure route tables for internet access
- 5) Launch an EC2 instance in the public subnet
- 6) Configure security groups to allow SSH (22), HTTP (80), and HTTPS (443)

EXPECTED OUTCOME

The EC2 instance in the public subnet will be able to access the internet, while resources in the private subnet will remain secure and isolated.

ASSIGNMENT 2

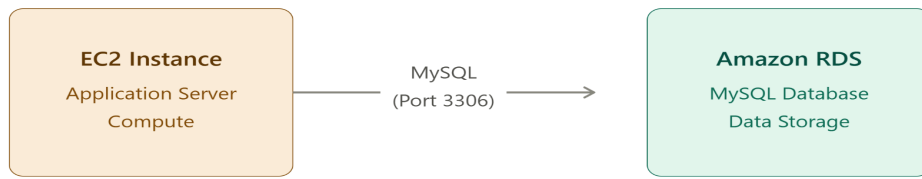
Create and Connect AWS RDS Database

OBJECTIVE

The objective of this assignment is to understand how to create and connect a database using Amazon RDS. RDS simplifies database management by handling backups, updates, and scaling automatically.

EXPLANATION

Amazon RDS (Relational Database Service) is a managed database service provided by AWS. It supports databases like MySQL, PostgreSQL, and others. It allows users to focus on application development without worrying about database maintenance.



EC2 connects to RDS over secure network connection

STEPS

- 1) [Launch an RDS MySQL instance](#)
- 2) [Select the same VPC as the EC2 instance](#)
- 3) [Configure database username and password](#)
- 4) [Allow inbound traffic on port 3306](#)
- 5) [Connect to the database using a MySQL client](#)

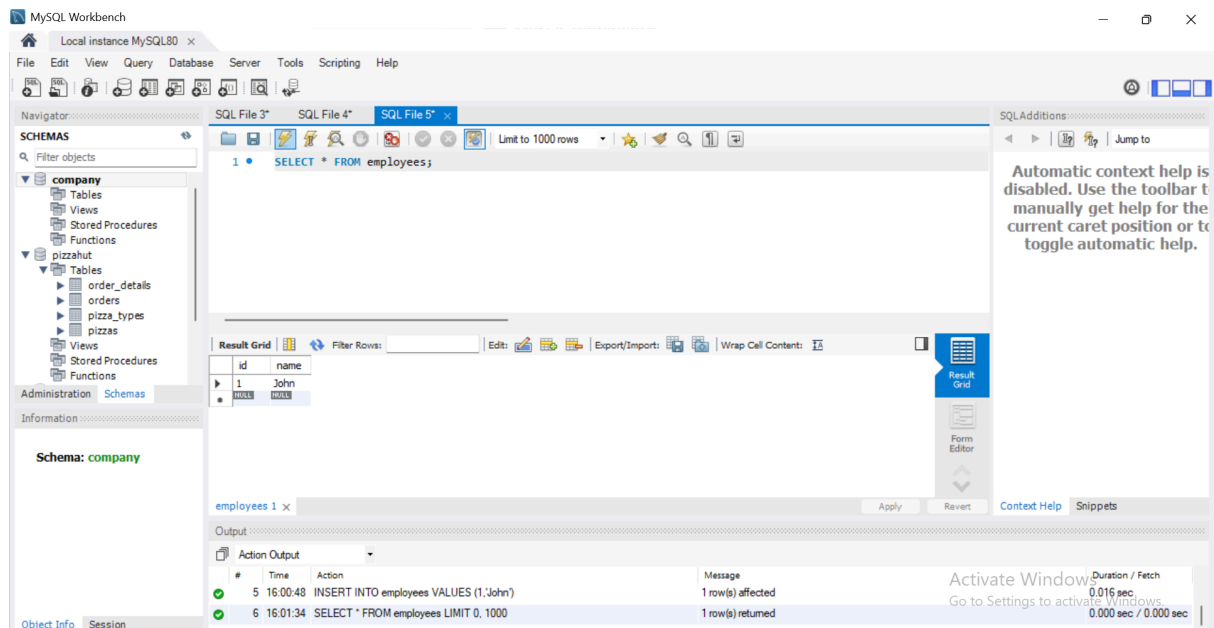
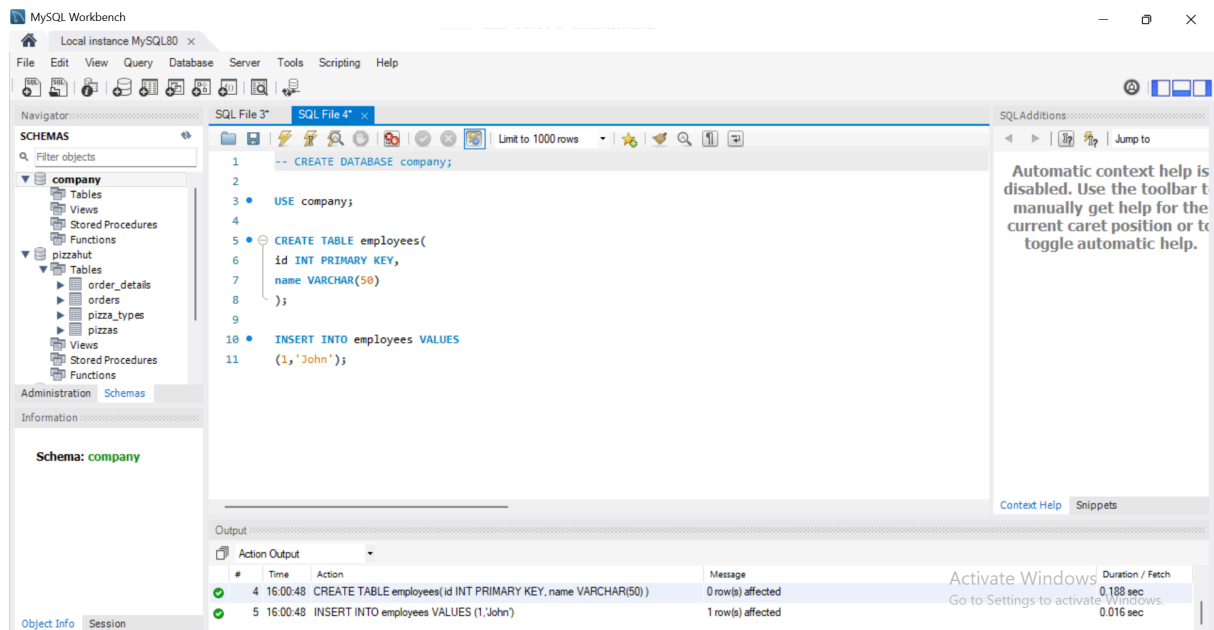
SQL IMPLEMENTATION

SQL Code

```
CREATE DATABASE company;
USE company;
CREATE TABLE employees (
  id INT PRIMARY KEY,
  name VARCHAR (50)
);
id INT PRIMARY KEY,
(1, 'John');
SELECT * FROM employees;
```

Output:

1 row inserted | Database and table created successfully



EXPECTED OUTPUT

The EC2 instance successfully connects to the RDS database and performs database operations like creating tables and inserting data.

MINI PROJECT

“Serverless Contact Form using Lambda & API Gateway”

OBJECTIVE

The objective of this project is to understand serverless computing using AWS Lambda and API Gateway. Serverless architecture allows applications to run without managing servers.

EXPLANATION

In serverless architecture, AWS Lambda executes code in response to events, and API Gateway acts as an entry point for client requests. This reduces infrastructure management and improves scalability.

WORKFLOW

1. The user fills out a contact form
2. API Gateway receives the request
3. AWS Lambda processes the request
4. Lambda returns a response to the user

LAMBDA CODE

```
def lambda_handler(event, context):  
    return {  
        "statusCode": 200,  
        "body": "Form submitted successfully"  
    }
```

OUTPUT

```
{
```

```
"statusCode": 200,  
"body": "Form submitted successfully"  
}
```

CONCLUSION

In this assignment, we learned how to create a Virtual Private Cloud (VPC) and manage networking components in AWS. We also explored Amazon RDS for database management and implemented basic SQL operations. Additionally, we understood serverless computing using AWS Lambda and API Gateway. This assignment helped in understanding core cloud concepts and their practical applications.